

5.12 Solution: By definition, torque is given by

$$\mathbf{N} = \int \mathbf{r} \times (\mathbf{j} \times \mathbf{B}) d^3x = \int [\mathbf{j}(\mathbf{r} \cdot \mathbf{B}) - \mathbf{B}(\mathbf{r} \cdot \mathbf{j})] d^3x.$$

Here, $\mathbf{j}$ is the current density on the inner loop with radius b , and $\mathbf{B}$ is the magnetic induction from the outer loop with radius a on the inner loop. It is easy to see that $\mathbf{r} \cdot \mathbf{j} = 0$ on the inner loop. Then

$$\mathbf{N} = \int \mathbf{j}(\mathbf{r} \cdot \mathbf{B}) d^3x = b \int B_r \mathbf{j} d^3x,$$

where, from Section 5.5, we have for a current loop on the xy plane,

$$\begin{aligned} B_r &= \frac{\mu_0 I a}{2r} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{r_{<}^{2n+1}}{r_{>}^{2n+2}} P_{2n+1}(\cos \theta) \\ &= \frac{\mu_0 I}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{b^{2n}}{a^{2n+1}} P_{2n+1}(\cos \theta), \end{aligned}$$

with $r_{<} = r = b$, and $r_{>} = a$.

In the coordinate system, it is not very easy to determine θ for the inner loop, as it makes an angle with the xy plane. We can rotate the whole system around the y -axis by an angle of α so that the inner loop now lies in the xy plane. The original z -axis now points in the $(\theta, \phi) = (\alpha, 0)$ direction, and the points on the inner loop can be parameterized by $(\theta', \phi') = (0, \phi')$. Then the angle γ between the original z -axis and any point on the inner loop becomes

$$\cos \gamma = \cos \alpha + \sin \alpha \cos \phi'.$$

By the summation formula, Eq. (3.68), we have

$$\begin{aligned} P_l(\cos \gamma) &= P_l(\cos \alpha) P_l(\cos \theta') + 2 \sum_{m=1}^l \frac{(l-m)!}{(l+m)!} P_l^m(\cos \alpha) P_l^m(\cos \theta') \cos[m(\phi - \phi')] \\ &= P_l(\cos \alpha) P_l(0) + 2 \sum_{m=1}^l \frac{(l-m)!}{(l+m)!} P_l^m(\cos \alpha) P_l^m(0) \cos(m\phi'). \end{aligned}$$

Also, the current density on the inner loop in the new coordinate system is

$$\mathbf{j} = -J_\phi \sin \phi' \hat{\mathbf{i}} + J_\phi \cos \phi' \hat{\mathbf{j}},$$

where

$$J_\phi = I \sin \theta' \delta(\cos \theta') \frac{\delta(r' - b)}{b}.$$

Then

$$\begin{aligned} \mathbf{N} &= b \int B_r \mathbf{j} d^3x = b \int r'^2 dr' \int_{-1}^1 d(\cos \theta') \int_0^{2\pi} d\phi' \left(-J_\phi \sin \phi' \hat{\mathbf{i}} + J_\phi \cos \phi' \hat{\mathbf{j}} \right) \\ &\quad \times \frac{\mu_0 I}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{b^{2n}}{a^{2n+1}} \end{aligned}$$

$$\times \left[P_{2n+1}(\cos \alpha) P_{2n+1}(0) + 2 \sum_{m=0}^{2n+1} \frac{(2n+1-m)!}{(2n+1+m)!} P_{2n+1}^m(\cos \alpha) P_{2n+1}^m(0) \cos(m\phi') \right].$$

We can see that the integration only has non-zero component in $\hat{\mathbf{j}}$ direction. Also the ϕ' integration is non-zero for $m = 1$ only. Noting $P_{2n+1}(0) = 0$, we finally have

$$\begin{aligned} \mathbf{N} &= \frac{\mu_0 \pi I I' b}{2} \cdot b \pi \hat{\mathbf{j}} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{b^{2n}}{a^{2n+1}} \frac{2(2n)!}{(2n+2)!} P_{2n+1}^1(\cos \alpha) P_{2n+1}^1(0) \\ &= \frac{\mu_0 \pi I I' b^2}{2a} \hat{\mathbf{j}} \sum_{n=0}^{\infty} \left(\frac{b}{a} \right)^{2n} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{1}{(n+1)(2n+1)} P_{2n+1}^1(\cos \alpha) \frac{(-1)^{n+1} \Gamma(n + \frac{3}{2})}{\Gamma(n+1) \Gamma(\frac{1}{2})}. \end{aligned}$$

Since

$$\frac{\Gamma(n + \frac{3}{2})}{\Gamma(n+1) \Gamma(\frac{1}{2})} = \frac{(2n+1)!!}{2^n (n+1)!},$$

then we can write the above expression as

$$\mathbf{N} = -\hat{\mathbf{j}} \frac{\mu_0 \pi I I' b^2}{2a} \sum_{n=0}^{\infty} \left(\frac{b}{a} \right)^{2n} \frac{n+1}{2n+1} \left[\frac{\Gamma(n + \frac{3}{2})}{\Gamma(n+1) \Gamma(\frac{1}{2})} \right]^2 P_{2n+1}^1(\cos \alpha).$$

Here we have assumed that the current on the inner loop is counter clockwise. There will be a sign flip with the direction of the current flow.